

Micromouse - Kit Overview

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

1 Introduction

This kit provides all the necessary components to be competitive for the Micromouse challenge. This document provides an overview of the components within the kit as well as the PCB wiring.

1.1 IMPORTANT NOTE

Do not solder the core components directly to the PCB. You must use the provided headers. If you solder components directly to the PCB and encounter a problem, we will not supply replacement parts.

2 Component Overview

2.1 Core components

These components are supplied initially. Replacements will not be provided unless you can prove the component was damaged when you received it.

BILL OF MATERIALS - Core Components			
COMPONENT	DESCRIPTION	MAUFACTURER	PART No.
XA1	Arduino Nano	Arduino	A000005
XA2	IMU	Core Electronics	018-MPU-6050
XA3	Dual Motor Driver Carrier, DRV8835 (for Motor-Driver)	Pololu	2135
XA4	OLED screen, White, 64x128 pixels	Core Electronics	CE09493
B1	Battery, Holder, 9V, PCB	Memory Protection Dev	BA9VPC
B1	Battery, 9V, Li-Ion	paleblue	9V-USB-C
A2, A3 & A4	TOF Distance Sensor (Lidar)	Pololu	2489
HARDWARE	Controller PCB		MDICP11A
HARDWARE	Battery PCB		MDIBP11A
HARDWARE	2x Micro Metal Gearmotor w/Encoder	DFROBOT	FIT083
HARDWARE	Wheel 32*7mm pair - White	Pololu	1088

Figure 1: Bill of Materials for core components.

2.2 Trolley components

These components can be found in a trolley within the mechatronics lab. Please only take the required amount. If you damage one of these components, you may replace it. If your team decides to make your own custom PCB, you are free and encouraged to take these components such that you do not solder a sensor directly to your PCB.

BILL OF MATERIALS - Controller PCB - Assembly 1			
COMPONENT	DESCRIPTION	MAUFACTURER	PART No.
A1D1	LED, 5mm, Green, Round (for Pwr On)		
A1D2	Diode, Schottky, 30V, 5A, 480mV (for Pwr)	Vishay	1N5817
HARDWARE	Standoff - M3 x 11mm, F-F (for IMU)	Würth Elektronik	970110365
HARDWARE	M3x6 SEMS Screw (for IMU)	Hobson Engineering	23P36FS
HARDWARE	Standoff - M2 x 12mm, F-F (for OLED)	Essentra Components	HTSN-M2-12-5-1
HARDWARE	M2x6 PH Screw (for OLED)	Hobson Engineering	21PS26
HARDWARE	M2 Spring Washer (for OLED)	Brighton Best	71PS2
A1J1	Connector, XT30, Male, PCB Mount, Right Angle, 15A	multicompPRO	MP011756
A1J10 & A1J11	Socket Head, Single Row, 15 way (for Nano)	Amphenol	76341-315LF
A1J12	Socket Head, Single Row, 8 way (for IMU)	Würth Elektronik	61300811821
A1J2, A1J6 & A1J7	Socket Head, Single Row, 2 way (for Spare AIO, 5V & 3.3V)	Würth Elektronik	61300211821
A1J3	Socket Head, Single Row, 6 way (for Spare DIO)	Würth Elektronik	61300611821
A1J4, A1J5 & A1J13	Socket Head, Single Row, 4 way (for I2C, GND & OLED)	Würth Elektronik	61300411821
A1J8, A1J9, A1J14-A1J16	Socket Head, Single Row, 7 way (for Motor-Driver & Lidars)	Würth Elektronik	61300711821
A1R1	Resistor - 330 Ohm, 0.6W, 1%, TC50 (for Batt Pwr LED)		
A1ST1 & A1ST2	Screw Terminals - 6 way	Buchanan	282834-6

Figure 2: Bill of Materials for the Controller PCB trolley components.

BILL OF MATERIALS - Battery PCB - Assembly 2			
COMPONENT	DESCRIPTION	MAUFACTURER	PART No.
A2D1	Diode, 30V, 5A, 480mV, Schottky	Vishay	SB530-E3/54
A2D2	LED, Red, 5mm, Round		
A2F1	Fuse, Holder, MINI, PCB, Horizontal	Littelfuse	015300072
A2F1	Fuse, 4A, MINI	Littelfuse	0297004.WXNV
A2J1	XT30, 15A, PCB, Female, Right Angle	RS PRO	8051699
A2R1	Resistor, 330Ohm, Metal Film, 0.6W, 50PPM	multicompPRO	MCMF006FF3300A50
A2SW1	Switch, SPDT, Right Angle	E-Switch	100SP1T2B4M6QE

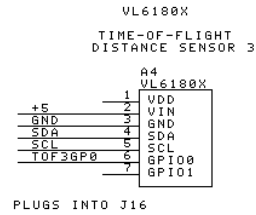
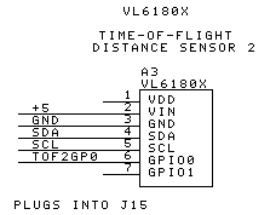
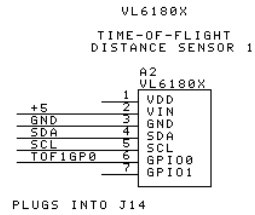
Figure 3: Bill of Materials for the battery PCB trolley components.

2.3 Battery

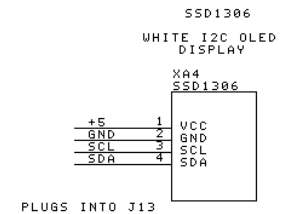
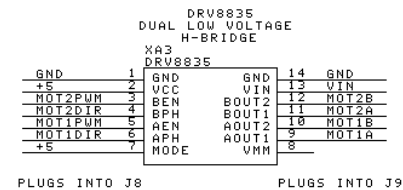
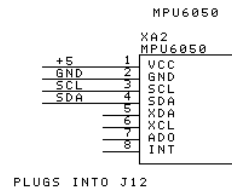
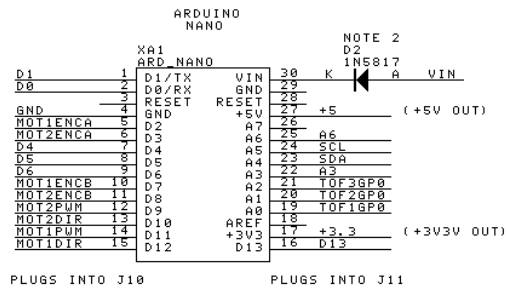
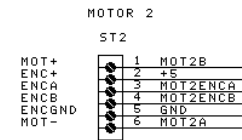
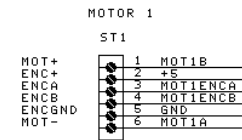
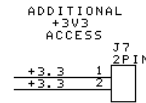
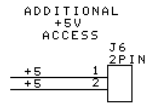
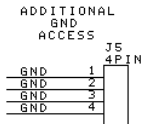
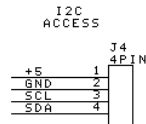
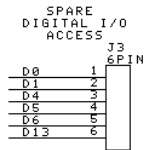
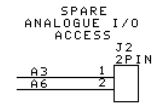
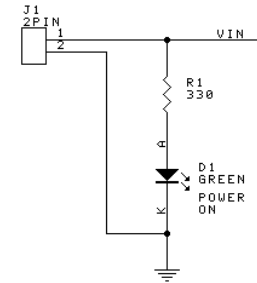
The supplied 9V batteries have a built-in resettable fuse to protect against short circuits. If you short-circuit the battery or it is not outputting the correct voltage, plug it into the charger to reset the fuse. USB-C chargers can be found in the mechatronics lab. It is your responsibility not to lose your misplaced battery.

3 Wiring information

Wiring information for the PCB has been supplied in the schematic attached to this document.



FROM J1 ON MDI BATTERY PCB
REFER TO THE FOLLOWING DRAWINGS:
MDIBS11A - MDI BATTERY PCB SCHEMATIC
MDILC11A - MDI LINK CABLE ASSEMBLY



- NOTES:
- ALL DRAWING NUMBERS REFER TO MECHATRONICS DRAWINGS UNLESS SPECIFIED OTHERWISE.
EXAMPLE: ABCDX12C/3A REFERS TO MECHATRONICS DRAWING NUMBER ABCD, X = (Schematic, Wiring diagram, Cable assembly or PCB), SHEET 1, VERSION 2, REVISION C, SHEET REFERENCE COLUMN 3, ROW A.
 - DIODE USED TO STOP ARDUINO FROM SENDING POWER OUT ON VIN WHEN CONNECTED TO USB.



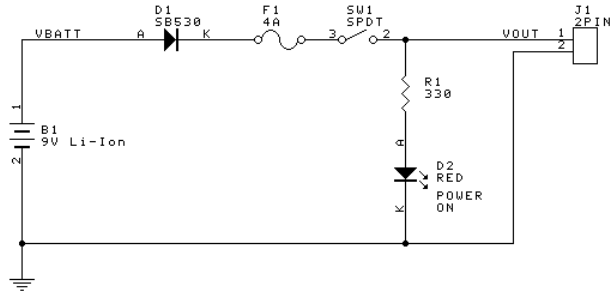
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2025



PCB NO: MDICP11A

Title MECHATRONICS - HTKN3100 MICRO DRONE INTERFACE CONTROLLER PCB - SCHEMATIC		
Size	Number	Revision
A3	MDICS11	A
Date: 28-APR 2025	Sheet 1 of 2	
File: MDICS11A/1	Drawn By: S.KÜHLE	

DRAWING PRODUCED BY: UNSW SYDNEY, ENGINEERING, MECHANICAL & MANUFACTURING ENGINEERING, MECHATRONICS
www.engineering.unsw.edu.au/mechanical-engineering



CONNECTS TO J1 ON MDI CONTROLLER PCB
REFER TO THE FOLLOWING DRAWINGS:
MDICS11A/7A - MDI CONTROLLER - SCHEMATIC
MDILC11A/3B - MDI LINK CABLE - CABLE ASSEMBLY

NOTES:
1. ALL DRAWING NUMBERS REFER TO MECHATRONICS DRAWINGS UNLESS SPECIFIED OTHERWISE.
EXAMPLE: ABCDX12C/3A REFERS TO MECHATRONICS DRAWING NUMBER ABCD, X = (Schematic,
Wiring diagram, Cable assembly or PCB), SHEET 1, VERSION 2, REVISION C, SHEET REFERENCE COLUMN 3, ROW A.



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PCB NO: MDIBP11A

Title MECHATRONICS - MTKN3100 MICRO DRONE INTERFACE BATTERY PCB - SCHEMATIC		
Size A3	Number MDIBS11	Revision C
Date: 5-MAY 2025		Sheet 1 of 2
File: MDIBS11C/1		Drawn By: S.KUHLE